Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years building **React** front ends, **Python** back ends, and **SQL** data pipelines, applying AI and algorithmic problem solving to large datasets, with cross-platform delivery and payments reconciliation experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant schemas on Azure SQL and SQL Server with **row-level security** for 500+ users and 10K+ records, cutting dashboard latency from 1.2s to 350ms via query optimization.
- Built SQL transformation and **reconciliation** pipelines across PostgreSQL, SQL Server, and cloud storage with incremental loads and data validation, running 5K+ monthly jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **AI-powered** platform with React and Flask, chaining GPT-4o, Claude, and Qwen-VL prompts for rubric generation, OCR, and grading, serving 150+ instructors and 10K+ daily requests.
- Translated grading workflows into a **concurrent** batch-processing design with secure PDF, DOCX, and ZIP ingestion, processing 1K+ files per batch and cutting feedback turnaround 60% for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Delivered **cross-platform** data-sharing software interoperating across Windows, Linux, and macOS, with automated VM executable builds, installs, and 20+ unit and integration test suites in nightly Jenkins pipelines.
- Redesigned AWS Glue and S3 **ETL** that normalized large JSON, CSV, and Parquet datasets from multiple sources, boosting pipeline efficiency 68% and improving API response times 30%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered 21 statistical features such as IP presence, URL length, and suspicious keywords across a **650K+ URL dataset**, translating a phishing and malware detection problem into a classification pipeline.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN models, reaching 96%+ **detection accuracy** and documenting actionable insights on phishing and malware trends.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated PCI-compliant multi-currency **payment** flows across cards, UPI, wallets, and PayPal with idempotent webhooks keeping booking and payment status consistent, cutting manual refund and dispute handling by 70%.
- Built hourly Databricks and Spark **analytics** over 50K+ booking and payment events, aggregating revenue, peak hours, and service demand for client reporting behind sub-200ms Python Flask APIs.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Full Stack: React, Next.js, React Native, Node.js, FastAPI, Flask, REST APIs
Data & Analytics: PostgreSQL, SQL Server, Oracle, Databricks, Spark, AWS Glue, ETL Pipelines, Data Validation, Reconciliation
AI & Algorithms: LLM Applications, Prompt Chaining, RAG, Feature Engineering, Random Forest, LSTM, Algorithm Design, Statistical Analysis
Platforms & DevOps: Windows, Linux, macOS, Cross-Platform Testing, AWS, Azure, Docker, CI/CD, Jenkins
Practices: Requirements Gathering, Technical Documentation, Agile/Scrum

Projects

Regulatory Reconciliation Assistant | *React, TypeScript, Python, FastAPI, PostgreSQL, Claude*

- Designed a React and FastAPI tool that reconciles ledger extracts against **regulatory report** line items in SQL, flags breaks by rule, and uses an LLM to draft documented explanations for analyst review.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, Kubernetes*

- Built a full-stack **fintech** platform in TypeScript, Next.js, and MongoDB with Plaid bank linking, advanced transaction filtering, and a token-based firewall across 1M+ transactions for 50K+ users.

Chat2SQL Code | *Java, TypeScript, SQL Server, Oracle, Docker*

- Built an AI-assisted SQL workspace on Java and TypeScript supporting 10+ databases including SQL Server and Oracle, generating schema-aware queries at 82% accuracy for **enterprise data** exploration.